

to exert greater control over their personal information, potentially enabling more stringent privacy settings and personalised data-sharing preferences. The implementation of technologies such as blockchain in decentralised networks can further secure data transactions and ensure transparency, while maintaining user anonymity and privacy.

Federated learning for security of AI models

Federated learning represents a paradigm shift in the development and deployment of AI models, premised on the decentralisation of data processing. This innovative approach is instrumental in enhancing the security and privacy of AI models in a hyper-connected digital ecosystem. The necessity of federated learning emerges primarily from the growing concerns around data privacy, security breaches, and the centralised nature of traditional data processing systems.

Using federated learning, data is fragmented in chunks and distributed to multiple locations or organisations where multiple machine learning models are created in Pickle (PKL) or any other format. These multiple PKL files are collected from different locations ensuring that the entire dataset is not given to one location or single organisation.

Federated learning introduces an additional layer of security by leveraging techniques such as differential privacy and secure multi-party computation. Differential privacy ensures that the shared updates do not allow for the reconstruction of the underlying data, effectively anonymising individual contributions to a degree that prevents potential misuse. These methodologies ensure that the collaborative training process across multiple decentralised nodes does not compromise the confidentiality of the data.

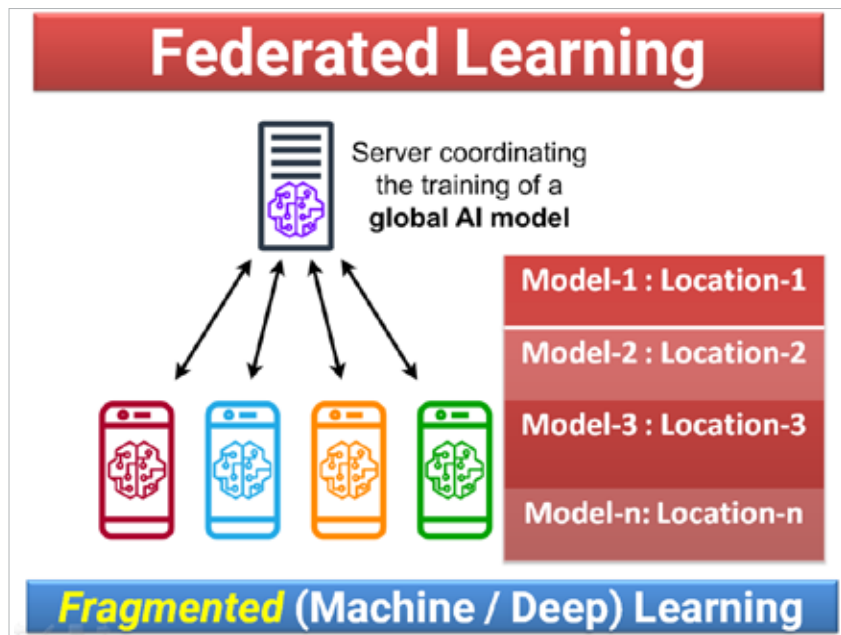


Figure 3: Diversification of models and data using federated learning

Patient ID	Age	Diabetic	Blood Pressure Level	Disease
1	45	0	120	0
2	60	1	140	1
3	35	0	110	0
4	50	1	130	1
5	70	0	150	1

Figure 4: Chunk of data taken from the entire dataset

Federated learning inherently supports a more robust model by aggregating a diverse array of data sources, each contributing unique insights that are reflective of varied user environments and behaviours. This diversity not only enriches the training process, resulting in more generalised and robust AI models, but also dilutes the risk of model poisoning or adversarial attacks, where malicious actors attempt to skew model performance by tampering with training data.

By decentralising the training process and ensuring that no single entity can monopolise data interpretation and subsequent AI application, federated learning promotes a balanced power distribution in the field of artificial intelligence. This not

only helps prevent data monopolies but also fosters a more competitive and diverse technological ecosystem, ultimately leading to more innovative and fair AI solutions.

Implementation of federated learning in Python

Let's suppose there is a dataset of one million records related to patients and we need to train a machine learning or deep learning model on it. Using federated learning, the entire dataset is divided into multiple chunks like `data_part1.csv`, `data_part2.csv`, `data_part3.csv`, and so on. In each chunk of data, there are different records or data files. Each of these data chunks or data parts is given to a different organisation to build an AI model, so that a single organisation cannot access or leak the entire dataset.